CMPE 246 Computer Engineering Design Studio

Project Proposal

Lab Group G2

Section L2A

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Abstract

A battery management system is a key component of any mobile electronic device. It ensures that the cells in the battery are evenly charged and that the battery is operated safely. Our project will create a framework for creating custom battery management systems, based on the Raspberry Pi 4. The Raspberry Pi will enable additional features, such as machine learning based state of charge estimation, cell health estimation and remote monitoring. The battery management hardware will be housed on a custom PCB and resources will be developed to enable other creators to develop their own PCBs for their personal projects. This document also outlines our timeline for this project, the distribution of tasks, the resources needed and our ideation process.

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Introduction

Inside of every mobile electronic device is an often overlooked battery management system. These systems can be found in every device that uses multiple cells as a power source, from smart phones to electric vehicles. A battery management system balances the voltages and levels of charge of all cells in a battery while monitoring the battery's output and temperature.

Making a battery management system (or BMS) for a specialized use case with a specific number of cells is a difficult task for hobbyists as completely assembled BMSs are difficult to come by. A battery management system consists mainly of an integrated circuit for monitoring and balancing cell voltages, a microcontroller to send commands to and receive data from this IC, as well as an additional chip to enable serial communication between the microcontroller and battery management circuit.

We aim to create an open-source framework for creating custom battery management systems for mobile electronic projects. Our design will integrate a Raspberry Pi 4 which will allow for users to send commands to the BMS remotely and receive data about the battery, such as charge level and power output, via their browser. The system will also include code leveraging machine learning to estimate the battery's state of charge and the health of the lithium-ion cells. The system we will produce for this project will contain five lithium-ion cells managed by a battery management IC on a custom PCB but the system architecture will be scalable to much larger battery sizes.

This project should allow hobbyists, student design teams and other engineers who need pre-production prototypes to easily create feature-complete battery management systems to support their own projects. Removing the hassle of battery design will allow engineers to focus on the design of their prototypes rather than supporting systems.

Team Composition

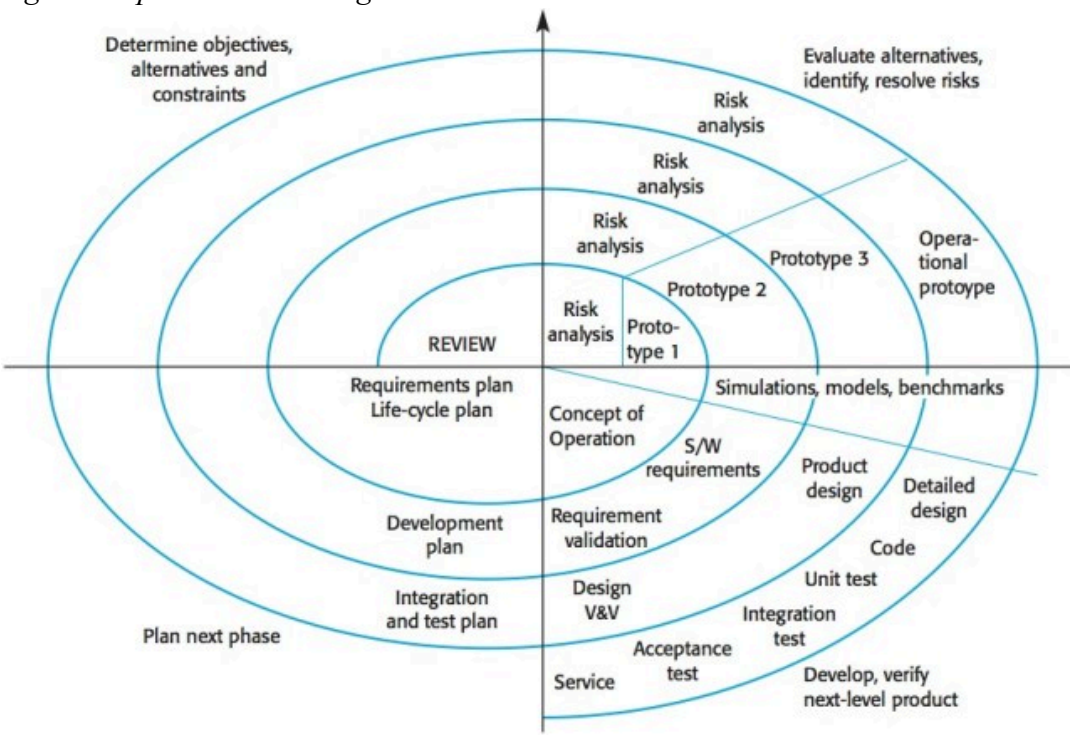
Table 1. Team Members and their Roles

Simon Houghton	C++ programming to control BMB, machine learning features
Lara Boljuncic	Design and software for the battery demonstration and screen
Maclean Ross-Howe	BMB hardware design
Sam Knight	Power circuit design and manufacturing

Ideation and Development

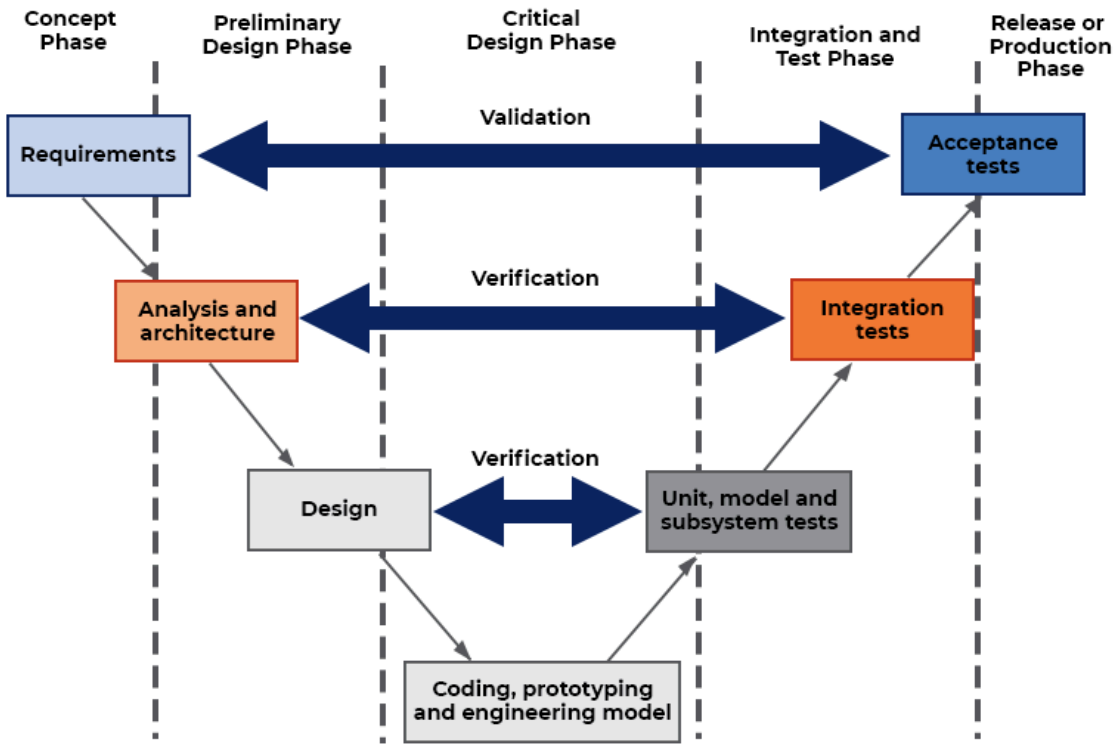
During the duration of the project, the team will follow two main design flows to ensure the project remains organized. The first design flow is the spiral model, seen in Figure 1. The team will start in the middle of the spiral, and continue following the path outwards, alternating between reviewing, prototyping, developing, and planning. The first stage will start with defining project requirements and goals. After this has been completed, the team will carry out ideation and begin initial prototypes. The team will then continue in the spiral of reviewing, planning, and prototyping, until the completion of the project.

Figure 1. Spiral Model Design Flow



The second design flow the team will follow is the V-model, seen in Figure 2. Similar to the spiral model, the team will start in the concept phase, defining requirements, before moving on to preliminary design. The project will proceed in stages along the "V", and go backwards for validation and verification of design before proceeding again. This cycle will continue for the duration of the project, finishing at release, which in this project is the presentation in Week 13. To ensure the team follows these design flows, weekly meetings will be set up to check in on progress and update the team timeline.

Figure 2. V-Model Design Flow



To begin ideation for this project, a need statement and project requirements were first defined. From there, three different ideation techniques are utilized to explore solutions and project topics: brainstorming, SCAMPER, and Heuristic Ideation Technique (HIT).

Need Statement:
"A way to address the use and integration of a Raspberry Pi 4 in the CMPE 246 course project that demonstrates embedded system design, hardware/software integration, and industry-ready technical competence through a real-world application."

Project Requirements:
Functions:
- Integration of a Raspberry Pi 4
- Develop knowledge of embedded systems and system architecture
- Engage in design flow and team collaboration
- Implement hardware/software integration and development
- Investigate trends related to embedded systems, communication network, security, Internet of Things (IoT) and Artificial Intelligence (AI)

Objectives:
- Minimize cost of hardware and implementation costs
- Maximize professional online presence through public documentation of the project
- Maximize constructive feedback from peers, instructors, and external audiences
- Maximize collaboration and communication

Constraints:

- Project must be fully completed by Week 13 for presentation
- Must integrate a Raspberry Pi 4 and operate within its limits
- Documentation of the design and development must be recorded on shared platforms

Brainstorming:

- Battery Management System → selected project
    - Monitor battery cell temperature, voltage, approximate cell health, and power a small motor
- Greenhouse
    - Light sensor, watering schedule, temperature monitoring
- Model Security System
    - Sensors for motion, open doors or windows, theft alarm
- House Automation
    - "Smart" home appliances, like thermostat, lights, door locks
- Environment Monitor
    - Light intensity, temperature, humidity
- Multi-Use LED Notification System
    - Uses light strips or small LED's to display information such as temperature, volume, sports game score
- Temperature Controlled Fan/HVAC System
    - Controls constant temperature in a space
- RFID Security System
    - Monitors and logs authorized access via RFID cards or tags.
- Automated Pet Feeder
    - Dispenses measured amounts based on a schedule or sensors

SCAMPER:

Battery Management System of 18650 Lithium Ion Cells

Substitute:

- Cell type
- Manual monitoring to sensor based
- Local to cloud storage
- Wired to wireless communication
- State of Charge (SOC) calculations to machine learning estimate

Combine:

- System to power a small motor
- System with a LCD display screen
- Battery charging and monitoring
- Raspberry Pi 4 with a custom PCB

Adapt:

- Powerbank to include sensors
- Open-source Raspberry Pi 4 existing projects

Modify:

- Size of packaging of system
- Use of system to power/monitor different objects
- LCD display to terminal display

- Existing battery systems to sensors based

Put to another use:

- Power another object
- Reuse components from existing systems

Eliminate:

- Sensors where values can be calculated from other existing sensors
- LCD display when values can be shown in other forms
- Wireless communication when wired works well
- Complicated mechanical design when focus is embedded system

Reverse:

- Design flow, where software is completed before hardware
- Order of information display
- Enclosure design before PCB design

HIT (Heuristic Ideation Technique):

Table. 2 Heuristic Ideation Technique Battery Management System and LED Notification System

	Status Indicator LED's	External Power Supply	Browser Interface	Wireless Communication
Cell Output Sensors	LED indicate cell output	Sensors on external power	Sensor data displayed on browser	Wirelessly communicate sensor outputs
Cell Balancing	LED indicate cell balancing	External power aids cell balancing	Cell balancing data displayed on browser	Cell balancing data wirelessly communicated
SOC/SOH Estimation	LED indicate SOC/SOH status	Machine learning estimations powered externally	Estimations calculated/displayed on browser	SOC/SOH estimations wirelessly communicated
LCD Display	LED analog display	LCD Display externally powered	_	LCD display received instructions wirelessly
Data Logging	LED indicate logging status	Data logging external	Data stored in browser	Data logged wirelessly stored

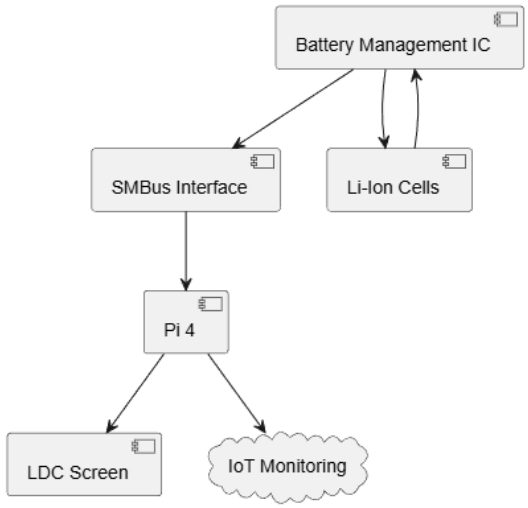
Solution Overview

The main purpose of the battery management system is to balance the voltages of each cell in the battery. This is accomplished by a Texas Instruments integrated circuit which automatically performs cell balancing during charging and discharging. This same IC can read the temperature of the battery at different points from a thermistor. The battery management chip can also receive commands to enter different modes and output data.

To connect the battery management IC to the Raspberry Pi 4 for data transmission, an accompanying chip is used for bidirectional serial communication over I2C. A logic level booster is used between the Pi and the I2C chip to convert between 3.3V and 5V communication. All of the components listed so far will be on a custom PCB along with the necessary additional circuitry.

The battery will consist of five 3.6V 18650 lithium-ion cells in series for a total pack voltage of 18V. Each cell is connected to the next via a ribbon of nickel sheet and each cell terminal will have a wire soldered to it for cell balancing, known as a current tap. The current taps will terminate in a connector that connects to the PCB. A 3D printed enclosure will safely house the cells and PCB.

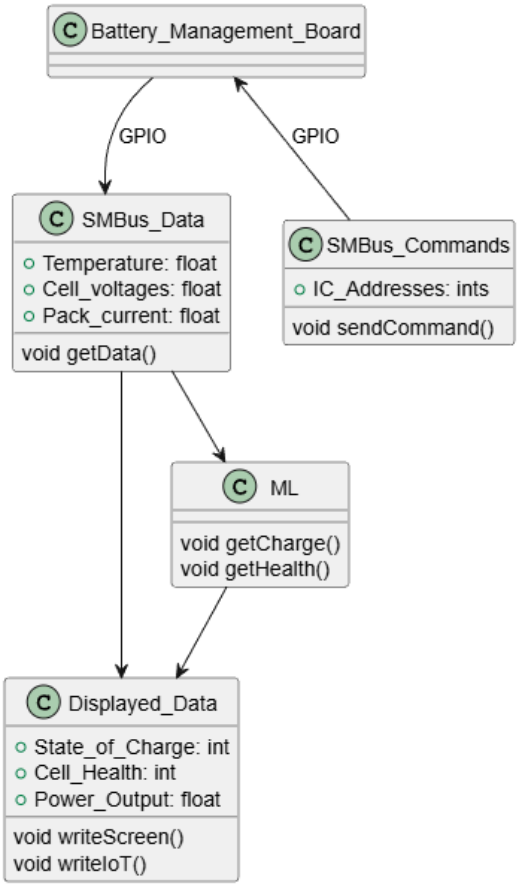
Figure 3. Flowchart of Interactions Between Physical Components



Note. This flowchart shows the electrical connections between the components in the battery management system. The connections between the Battery Management IC, SMBus Interface and Pi 4 are bi-directional.

On the Raspberry Pi, a program written in C++ will poll the data it receives from the battery management board via its GPIO pins. On startup, this program will tell the Texas Instruments chip which mode to enter and then will use the XGBoost inference API to estimate the battery's state of charge and the health of the cells at a low polling rate. The program will then output the data to an attached LCD screen and to a database to be read by a Python script. The Python script will host a local web server that can display the time series data in a graph.

Figure 4. Flowchart of Data Processing and Movement



Note. The variable and function names are only meant to represent what data is being obtained and how it is being processed. Battery_Management_Board represents the physical PCB while the other boxes represent components of the software.

All of the software and PCB designs we create will be available on GitHub for other creators to base their own BMSs off of. The code to send, receive and process data will be scalable to much larger and higher voltage batteries. Additionally, the capacity of the battery can be greatly increased by adding more cells in parallel without any change to the project whatsoever.

Required Resources

This project will require the materials outlined in Table 2. The 18650 lithium-ion cells and the nickel sheet will be sourced from the UBCO Motorsports Engineering Design Team. The wiring, DC to DC power supply, and display will be purchased online.

Table 3. Bill of Materials

Raspberry Pi 4 ×1
18650 lithium-ion cell ×5
Custom battery management PCB ×1
Nickel sheet ×1
18 gauge wire spool ×1
Male to female jumper wire ×2
Small LCD display with connector ×1
DC to DC power supply ×1

The custom battery management PCB is a complex part which requires many components. These will be sourced from JLC PCB, who will also manufacture the board.

Table 4. Bill of Materials for Custom Battery Management PCB (Active Components)

BQ76920 Battery Monitor ×1
BQ78350-R1A Battery Management Controller ×1

In addition to these physical components, the project will make use of GitHub for documentation and version control, Visual Studio Code for programming in C++ and the XGBoost C++ API to implement machine learning features. Data will be logged to InfluxDB and FastAPI will be used to host a local web server (separate from the C++ program) that can display realtime data over the local network. Grafana will be used as a frontend to view battery data remotely. If the complexity of the InfluxDM and Grafana setup is too high given the project timeline, the C++ program will write to a JSON file and a static HTML page will show its contents in a browser on the local network.

Timeline

To ensure the team completes the project by Week 13, a Gantt Chart is constructed to monitor tasks and status of completion. As the project continues, the Gantt Chart will be updated to match the current progress. This will help the team stay on track with the dependencies feature, making it clear which tasks must be completed before another. Since the Gantt Chart is generated in Notion, another view of project tasks can be seen in the Kanban Board to understand the status of each task. The Gantt Chart and Kanban Board can be seen in Figure 5 and Figure 6, respectively.

Figure 5. Gantt Chart in Notion

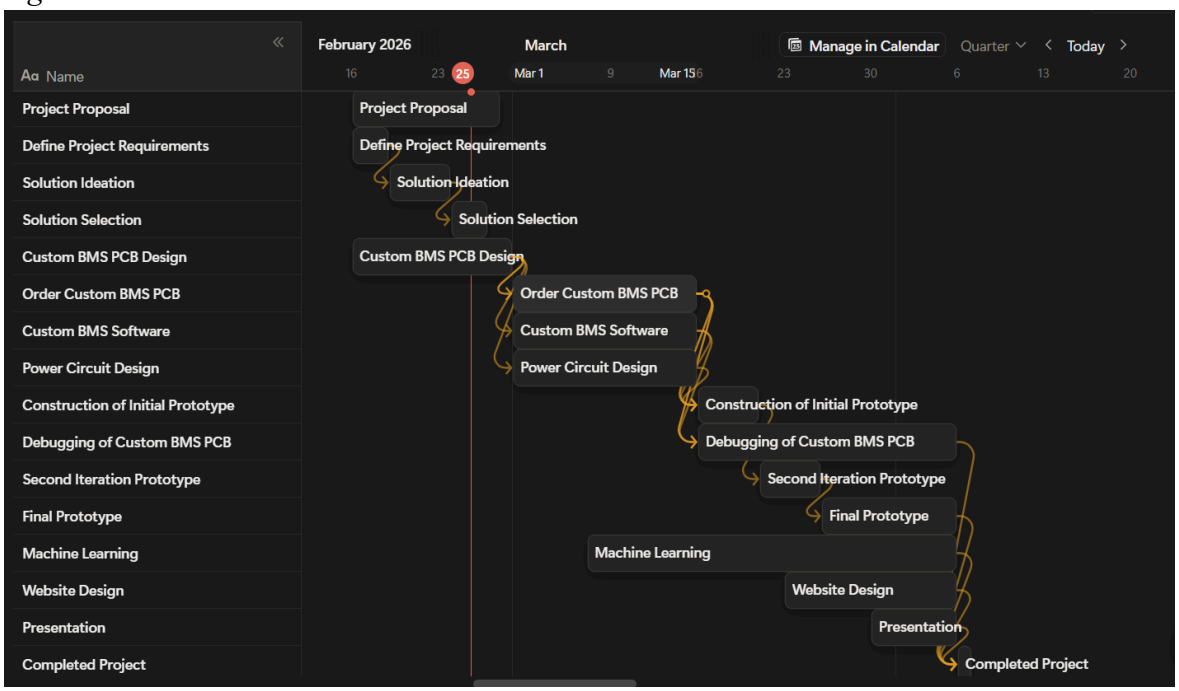
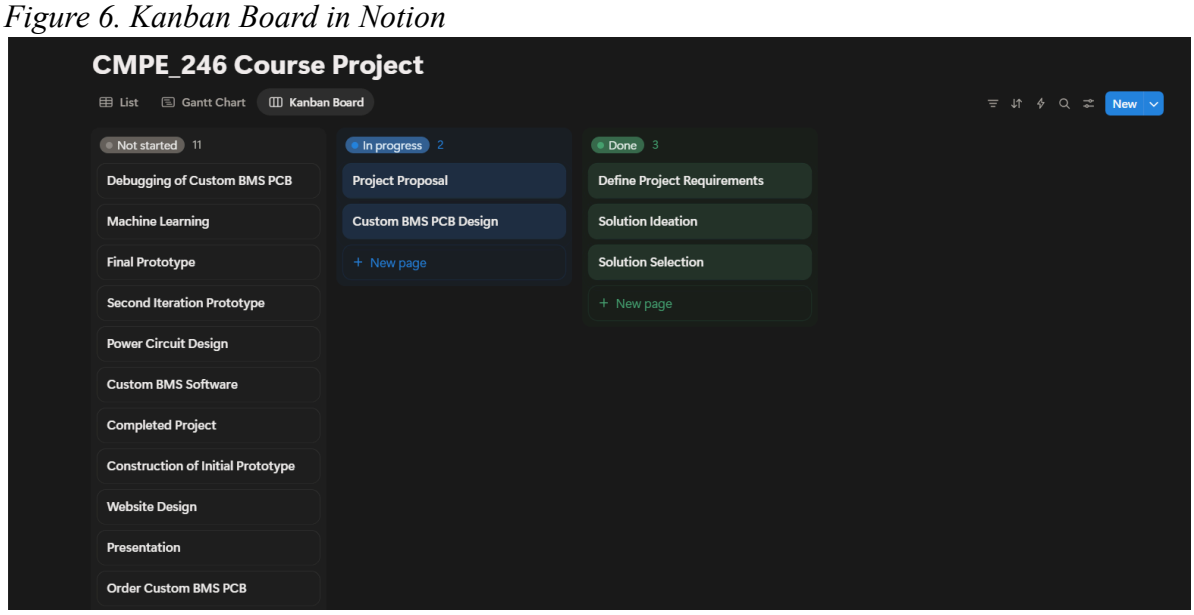


Figure 6. Kanban Board in Notion



Reflection

The result of our project will be a battery management system (BMS). The purpose of this project is to address the risks associated with charging lithium ion battery packs. Lithium ion batteries are sensitive to overcharging, high currents, and high temperatures and each of these can lead to cell degradation. This project aims to design a BMS that balances cell charging as well as protects the system from overcharging, short circuit currents, and excessive heating. In addition, the BMS will communicate with the Raspberry Pi to display voltage levels and temperature on a local web server.

A critical part of the project design is ensuring that the solution can fix the real world problem we aim to solve, and incorporates the use of Raspberry Pi and functions as an embedded system. As there is already existing design information for battery management systems we chose to add the additional functionality of using machine learning to estimate cell levels and display battery data on a web server. These functions will improve upon existing designs and will allow us to gain a better understanding of IoT, machine learning, and system architecture while utilizing a Raspberry Pi.

While developing the project proposal, our team faced several challenges related to scope and the complexity of our solution. Our main challenge was planning the software layers of the project such as how the BMB board will communicate with the Pi, and the necessary requirements to store data and display it on a local web server. To overcome these challenges, our team had to conduct research to develop a strong understanding of how the software layer works and discover various useful interfaces and modules that will help us achieve our goals such as XGBoost and InFluxDB.

The ideation phase of the project planning process was very effective and we were able to think of many possible projects as a team. Each member of the group had ideas for a project that we could make while including the use of Raspberry Pi in each option. Ultimately, we chose the battery management system because it offers complexity that will allow us to learn more about the integration of hardware and software, and embedded systems as a whole. Additionally, our chosen project is an option that all group members agreed upon, and had tasks that could be well defined and assigned to each group member.

This project has a strong mix of hardware and software requirements, and as such, will present several major challenges for our team. The major challenges that we expect to face are ordering and shipment time of the BMB board, learning and incorporating machine learning, and implementing effective I2C communication between the Raspberry Pi and the BMB board. Our strategy to address these challenges is to create a structured timeline that highlights critical tasks, maintain proper communication within the team, constantly re-evaluate the project progress, and spend time on additional learning outside of class time. As the BMB board is a critical task, we plan to complete the design and ordering of the board as soon as possible to avoid any subsequent tasks being delayed. Additionally, our team plans to meet at least once a week to review project progress and determine our next steps. These weekly meetings will also help the team review whether each function can be implemented within the project timeline and determine whether the overall project plan will require any adjustments. To address the challenges of implementing some difficult functions in our project, the team as a whole has committed to spending additional

time outside of the classes and labs to do additional learning so that these goals can be accomplished. We have also identified alternative options for functions such as the machine learning within our project plan to further mitigate these risks. By identifying these anticipated risks and developing mitigation strategies, our team is confident that we will be able to complete the project within the given timeline.

References

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